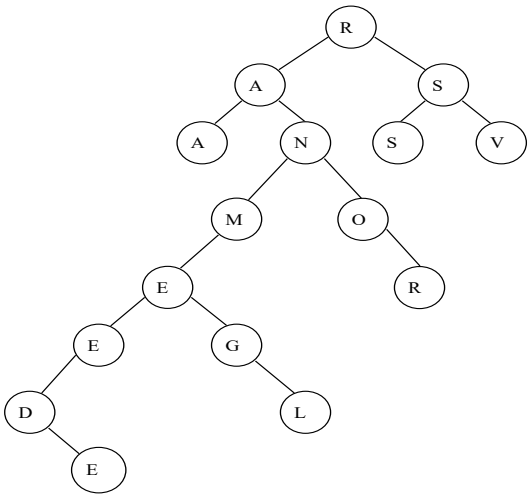
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1. **00-01 C4 Data Structures**

   The tree is formed as follows: 7



2. **00-01 C4 Data Structures**

   In order, the letters popped from the queue are: S E C R E T

3. **01-02 C4 Data Structures**

   The following is the result after each operation: D
   B, BI, BIN, IN, IND, ND, NDS, DS
   The next item to be popped would be the D

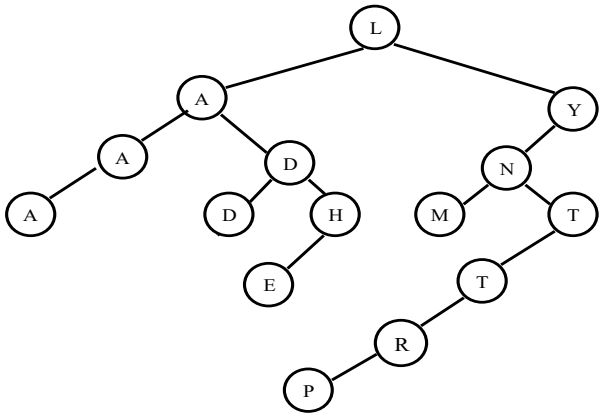
4. **01-02 C4 Data Structures**

   Vulcan becomes the right child of VENUS. 30
   The values to be added are: 2+3+3+3+3+4+4+4+4 = 30

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5. **02-03 C4 Data Structures**

The tree is formed as follows: 6

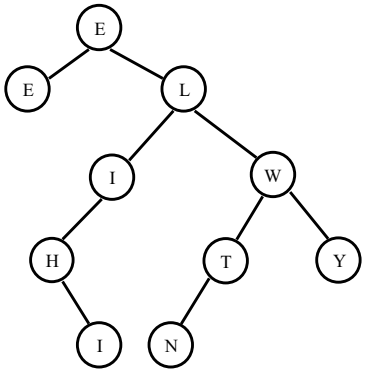


6. **02-03 C4 Data Structures**

The result up to each POP is as follows: SNWH
SNO, SN, SNWW, SNW, SNWHIT, SNWH, SNWHE, SNWH

7. **03-04 C4 Data Structures**

The tree is formed as follows: 3



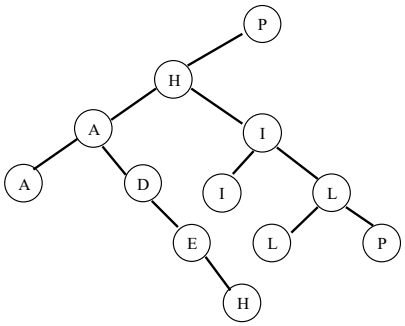
8. **03-04 C4 Data Structures**

The queue forms as follows: IL
CH, H, HIC, C, CAG, EMPTY, OI, I, IL

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9. **04-05 C4 Data Structures**

The tree is formed as follows: 5

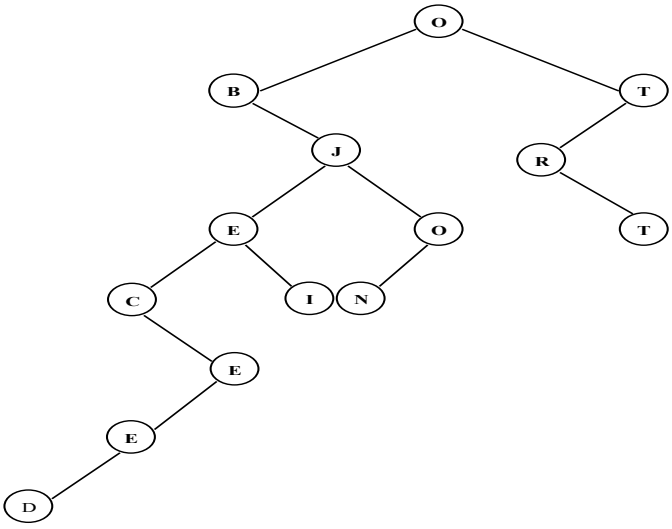


10. **04-05 C4 Data Structures**

The queue operates using the principal of FIFO. The contents of the queue changes as follows: LIB, IBE, ERT, RTY R

11. **05-06 C4 Data Structures**

The tree is formed as follows: 7



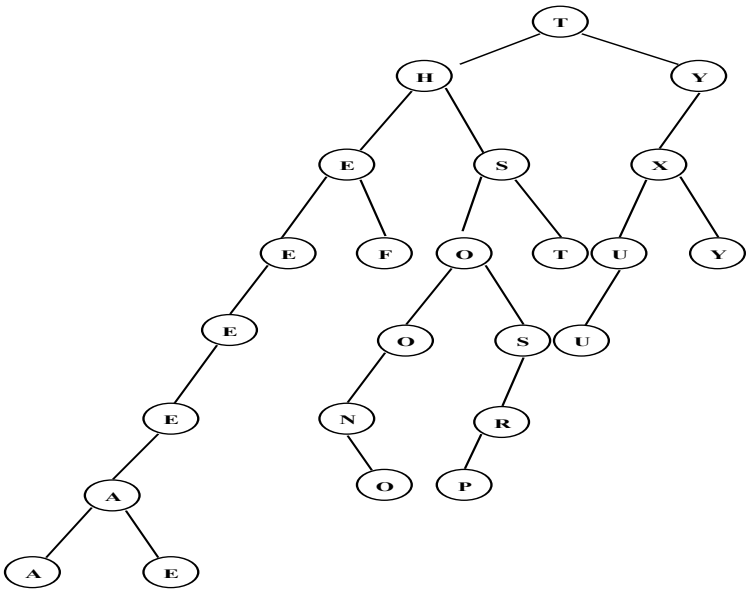
12. **05-06 C4 Data Structures**

The queue is constructed using FIFO as follows: A
INT, NT, NTE, TE, TERME, RME, RMED, ED, EDIA,
DIA, DIATE, ATE.
The next item to be removed would be A.

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13. **06-07 C4 Data Structures**

The tree is formed as follows: 7

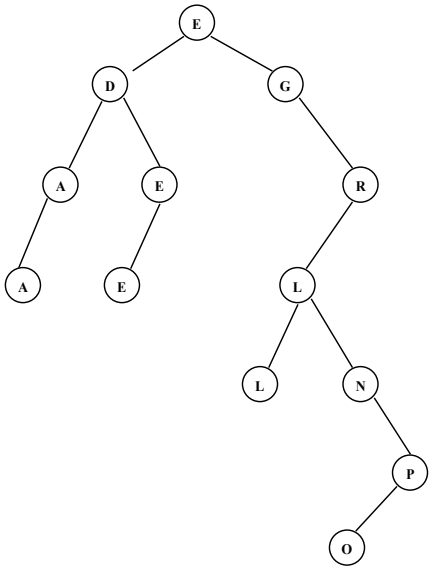


14. **06-07 C4 Data Structures**

The queue is constructed as follows: A
LON, OLE, ELOST, LOST, TSOL, SOL, LOSAR, OSAR, RASO, ASO. The next element POPPED is A.

15. **07-08 C4 Data Structures**

The tree is formed as follows: E, D, L (any order)



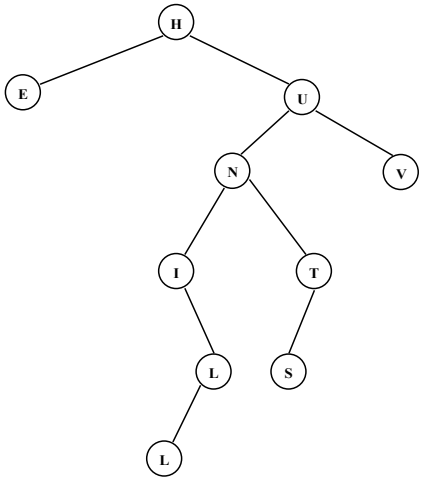
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16. **07-08 C4 Data Structures**

The queue is constructed as follows: E
THE, HE, HERA, RA, RAVE, AVE, AVEN, EN
The next popped item is E since queues are FIFO.

17. **08-09 C4 Data Structures**

The path length is found by adding 0 + 2*1 + 2*2 + 2*3 + 2*4 + 1*5 = 25. 25

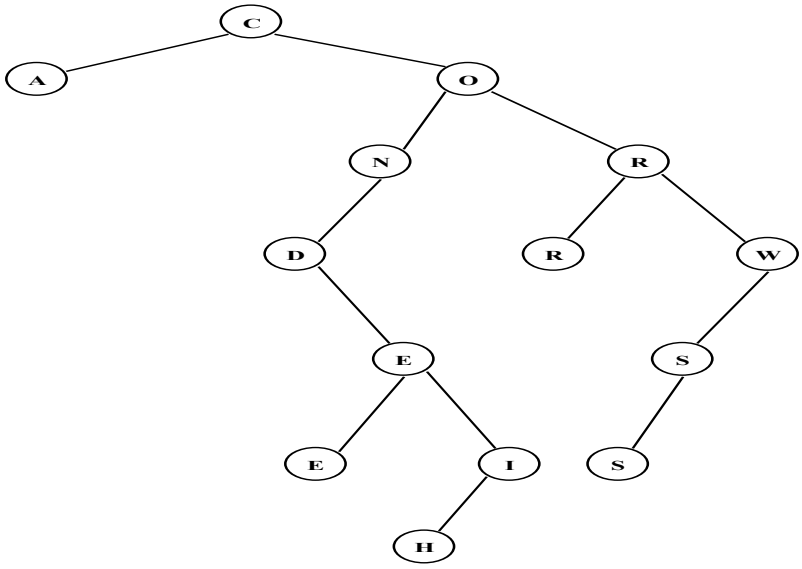


18. **08-09 C4 Data Structures**

The stack is constructed as follows: R
TRE, ERT, ER, ERAS, SARE, SA, SAUR, RUAS, RUASE, RUAS, SAUR
The next popped item is R since stacks are LIFO.

19. **09-10 C4 Data Structures**

The tree is formed as follows: 6



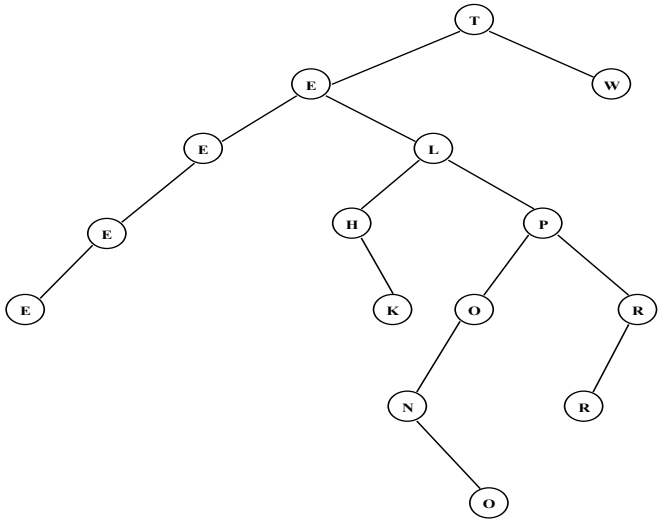
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20. **09-10 C4 Data Structures**

The STACK is constructed as follows: F
A, AB, ABC, ABCD, CDAB, CDA, CDAE, CDAEF, AEFCD, AEFC, AEF
The next popped item is F since queues are LIFO.

21. **10-11 C4 Data Structures**

The tree is formed as follows: 6

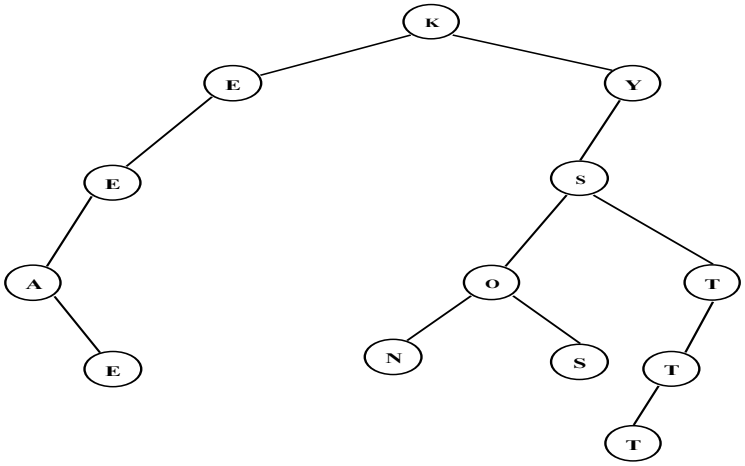


22. **10-11 C4 Data Structures**

The queue/stack is constructed as follows: N
PL, L, LA, L, LN, NN, N, NES
The next popped item is N since queues are FIFO.

23. **11-12 C4 Data Structures**

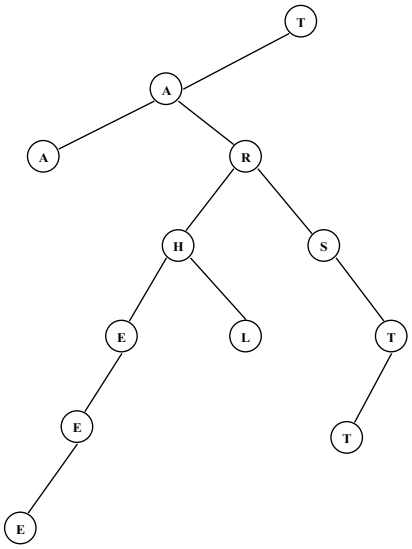
There are 6 nodes with exactly one child: A, E, E, S, T, T 6



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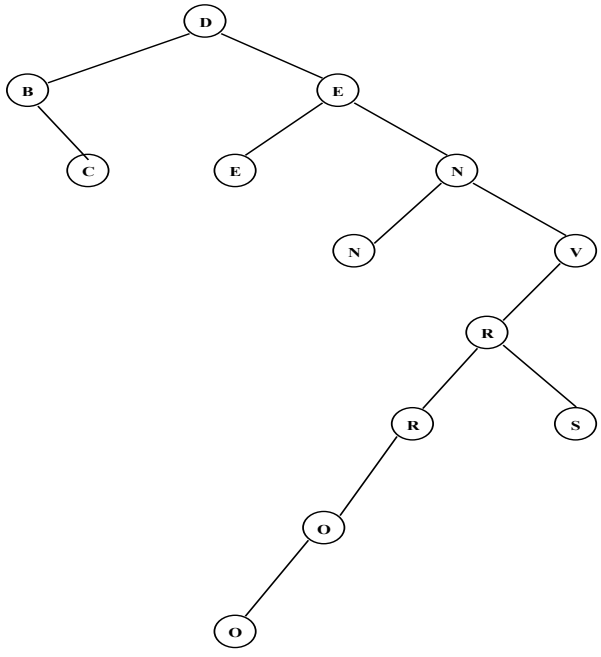
24. **11-12 C4 Data Structures**
The stack is constructed as follows: S
G, GE, GET, GE, G, GY, GYS, GYSB, GYSBU,
GYSB, GYSBR, GYSBRG, GYSBR, GYSB, GYS.
The next popped item is S since stacks are LIFO.

25. **12-13 C4 Data Structures**
The tree is formed as follows: 6



26. **12-13 C4 Data Structures**
The stack is constructed as follows: I
R, RA, RAL, RALE, RAL, RA, RAI, RAIG, RAIGH, RAIG,
RAIGD, RAIGDU, RAIGD, RAIG, RAIGR, RAIGRH, RAIGR,
RAIGRM, RAIGR, RAIG, RAI.
The next popped item is I since stacks are LIFO.

27. **13-14 C4 Data Structures**
The tree has 4 nodes with only one child: B, V, R, and O. 4



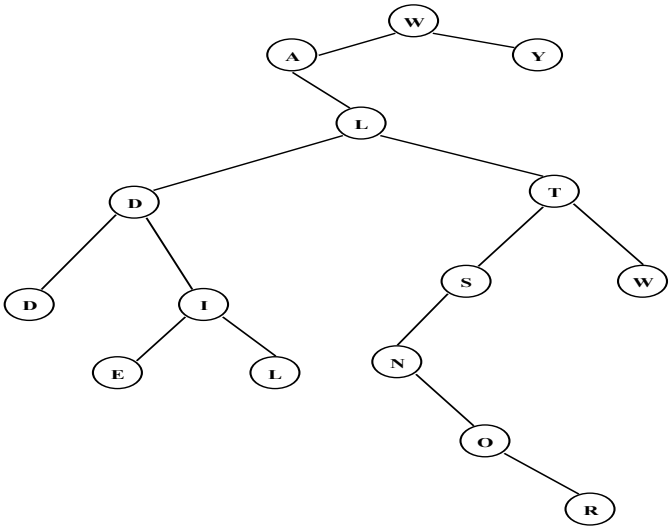
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28. **13-14 C4 Data Structures**

The queue is constructed as follows: P
P, PI, PIK, IK, K, KE, KES, KESP, ESP, ESPE, SPE, PE,
PEA, PEAK. The next popped item is P since stacks are FIFO.

29. **14-15 C4 Data Structures**

The tree is formed as follows: 7



30. **14-15 C4 Data Structures**

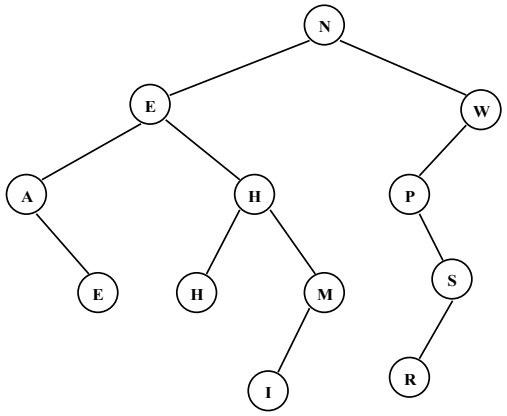
U

Queue:	SUN
Stack:	SUNSHI
	IHSNUSNE
Queue:	IHSNUSNESTATE
	SNUSNESTATE
Stack:	SNUSNESTA
	ATSENSU

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31. **15-16 C4 Data Structures**

The tree is formed as follows: 4



32. **15-16 C4 Data Structures**

The stack is constructed as follows: N
C, CO, C, CN, CNC, CN, CNO, CNOR, CNO, CN, CND, CNDN,
CND, CNDH, CND, CN. The next popped item is N since stacks are LIFO.